

Answer Key & Teacher Notes

Session 1 – Writing Numerical Expressions • Session 2 – Order of Operations with Powers

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Writing Numerical Expressions

TALK PROMPTS

- How do you decide whether a story needs you to add, to multiply, or to do both?
- In Tickets and Programs, why do the two products get added instead of multiplied together? Point to the words that tell you.
- Why does a numerical expression have no equal sign?

WATCH FOR

- Adding every number in the story instead of multiplying equal groups ($3 + 16 - 5$ for 3 bags of 16 crackers, minus 5 eaten).
- Writing “15 less than 40” as $15 - 40$.
- Leaving out parentheses when a whole sum or total must be found first ($100 - 3 \times 12 + 2 \times 20$).

PARTNER TASK

Field Trip and Tickets. One partner writes $20 \times 13 + 25$ for \$13 per student and a \$25 bus fee; the other evaluates it (\$285). Switch roles with 5 tickets at \$8 and 4 programs at \$5 ($5 \times 8 + 4 \times 5 = \60).

SESSION 2 • TEACHER NOTES

Order of Operations with Powers

TALK PROMPTS

- Why do mathematicians agree on one order instead of letting each person choose?
- In $40 - (6 + 2)^2 \div 4$, the division comes before the subtraction. Which rule puts it there?

WATCH FOR

- Working strictly left to right and adding before multiplying ($8 + 4 \times 3^2$ as 12×9).
- Applying an exponent to a product (2×5^2 as 10^2).
- Thinking multiplication always comes before division, or addition before subtraction.

PARTNER TASK

Same Numbers, Two Values. One partner evaluates $20 - 3 \times 4$ (8); the other evaluates $(20 - 3) \times 4$ (68). Then switch roles with $40 - (6 + 2)^2 \div 4$ (24) and $7 \times 2^3 - 9$ (47), naming each step aloud.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.EE.A.2	1	2	1	1	2	2	3	2	3	1	1
S1 • Form B	6.EE.A.2	1	2	1	1	2	2	3	2	3	1	1
S2 • Form A	6.EE.A.2c	1	2	1	1	2	3	2	2	3	1	1
S2 • Form B	6.EE.A.2c	1	2	1	1	2	3	2	2	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

numerical expression: Numbers and operations with no **equal** sign.

evaluate: Find the one **number** an expression equals.

product: The answer when you **multiply**.

- Equal groups ("each," "per") in a story usually mean **multiply**.
- A total of different groups means **add** the products.
- "Less than" flips the order: 5 less than 12 is **12 - 5**.

Example: Shirts are equal groups: 6×7 . Add the one hat: $6 \times 7 + 5$. Evaluate: $42 + 5 = 47$. Answer: $6 \times 7 + 5 = \$47$

YOU TRY

1. a. $6 \times 4 + 2$ b. $35 - 10$
2. $3 \times 5 + 4 = 15 + 4 = \19

EXIT TICKET

1. a. $30 - 9$ b. $3 \times (4 + 5)$
2. $6 \times 2 + 3 = 12 + 3 = \15
3. **A** ($4 \times 7 - 5$)

PRACTICE FORM A

1. a. $6 \times 3 + 4$ b. $75 - 20$ c. $3 \times (5 + 9)$ d. $48 \div 8 + 7$
2. a. $5 \times 12 + 4 \times 3 = 60 + 12 = \72 b. $10^2 = 100$ notebooks
3. $20 \times 13 + 25 = 260 + 25 = \285
4. $15^2 = 15 \times 15 = 225$ plants
5. Leo is wrong. "Each" means equal groups, so multiply: $3 \times 16 - 5 = 48 - 5 = 43$ crackers.
6. $12^3 = 12 \times 12 \times 12 = 1,728$ cubes
7. a. $3 \times 12 + 2 \times 20 (= \$76)$ b. $100 - (3 \times 12 + 2 \times 20) = 100 - 76 = \24 c. The parentheses show that the whole cost is found first, then subtracted from \$100.
8. $4 \times 9 + 3 \times 6 = 36 + 18 = \54
9. The 4 snacks at \$3 each belong in a product: 3×4 . The poster is added once. Correct: $3 \times 4 + 10 = 12 + 10 = \22 .
10. **A** ($5 \times 4 + 3$)
11. **B** ($40 - 15$)

Self-check word PRODUCT

PRACTICE FORM B

1. a. $7 \times 2 + 6$ b. $60 - 18$ c. $2 \times (3 + 7)$ d. $56 \div 7 - 3$
2. a. $6 \times 11 + 3 \times 8 = 66 + 24 = \90 b. $9^2 = 81$ tiles
3. $25 \times 12 + 30 = 300 + 30 = \330
4. $14^2 = 14 \times 14 = 196$ plants
5. Mia is wrong. "Each" means equal groups, so multiply: $4 \times 15 - 6 = 60 - 6 = 54$ pretzels.
6. $10^3 = 10 \times 10 \times 10 = 1,000$ cubes
7. a. $4 \times 10 + 3 \times 15 (= \$85)$ b. $100 - (4 \times 10 + 3 \times 15) = 100 - 85 = \15 c. The parentheses show that the whole cost is found first, then subtracted from \$100.
8. $5 \times 8 + 4 \times 5 = 40 + 20 = \60
9. 6 trays of 8 is equal groups, so 6 and 8 belong in a product. Correct: $6 \times 8 + 9 = 48 + 9 = 57$ seedlings.
10. **B** ($2 \times (8 + 2)$)
11. **A** (do first)

Self-check word NUMBERS

GUIDED NOTES

order of operations: The rules for which **step** to do first, next and last.

grouping symbols: Parentheses or brackets; do the work **inside** them first.

left to right: Do partner steps ($\times$ and $\div$, or $+$ and $-$) in the order they **appear**.

- First do work inside **grouping** symbols, then evaluate **powers**.
- Next multiply and divide from **left** to right.
- Last, add and subtract from left to **right**.

Example: Parentheses: $2 + 3 = 5$, so $40 - 5^2 \div 5$. Power: $5^2 = 25$, so $40 - 25 \div 5$. Divide, then subtract: $40 - 5$. Answer: $40 - (2 + 3)^2 \div 5 = 35$

YOU TRY

1. a. 2 b. 16
2. $2^3 = 8$; $8 \times 3 = 24$; $6 + 24 = 30$

EXIT TICKET

1. a. 14 b. 30
2. $2^2 = 4$; $4 \times 3 = 12$; $20 - 12 = 8$
3. **A** (32)

PRACTICE FORM A

1. a. 8 b. 68 c. 13 d. 54 e. 12 f. 11
2. a. $3^2 = 9$ b. $4 \times 9 = 36$ c. $8 + 36 = 44$; $44 - 10 = 34$
d. Left to right adds $8 + 4$ first: $12 \times 9 - 10 = 98$.
Multiplication must come before addition.
3. $6^2 \div 4 + 13 = 36 \div 4 + 13 = 9 + 13 = 22$
4. $5^2 + 8 \times 6 \div 2 = 25 + 48 \div 2 = 25 + 24 = 49$
5. Dara added $2 + 3$ first. Correct: $4^2 = 16$, $3 \times 16 = 48$, $2 + 48 = 50$.
6. Sample: $(24 \div 4 + 2) \times 3 = 8 \times 3 = 24$, or $24 \div (4 + 2 \times 3) = 24 \div 10 = 2.4$. (Note: $24 \div (4 + 2) \times 3$ is still 12.)
7. Kwame multiplied 4×3 before squaring. Correct: $3^2 = 9$, $4 \times 9 = 36$, $36 + 6 = 42$.
8. a. $4 + 12 = 16$ b. $6 \times 6 = 36$ c. Without parentheses, multiply first; the parentheses make you add first.
9. Powers come before multiplication, and the exponent applies only to 5. Correct: $5^2 = 25$, $2 \times 25 = 50$, $10 + 50 = 60$.
10. **B** (20)
11. **A** (before $\times$ and $\div$)

Self-check word **POWERS**

PRACTICE FORM B

1. a. 10 b. 130 c. 21 d. 24 e. 20 f. 17
2. a. $4^2 = 16$ b. $3 \times 16 = 48$ c. $7 + 48 = 55$; $55 - 12 = 43$
d. Elena adds $7 + 3$ first: $10 \times 16 - 12 = 148$.
Multiplication must come before addition.
3. $6^2 \div 4 + 10 = 36 \div 4 + 10 = 9 + 10 = 19$
4. $6^2 + 9 \times 6 \div 3 = 36 + 54 \div 3 = 36 + 18 = 54$
5. Theo added $3 + 2$ first. Correct: $5^2 = 25$, $2 \times 25 = 50$, $3 + 50 = 53$.
6. Sample: $(40 \div 5 + 3) \times 2 = 11 \times 2 = 22$, or $40 \div (5 + 3) \times 2 = 5 \times 2 = 10$.
7. Rosa multiplied 2×3 before cubing. Correct: $3^3 = 27$, $2 \times 27 = 54$, $54 + 4 = 58$.
8. a. $5 + 21 = 26$ b. $8 \times 7 = 56$ c. Without parentheses, multiply first; the parentheses make you add first.
9. Powers come before multiplication, and the exponent applies only to 4. Correct: $4^2 = 16$, $3 \times 16 = 48$, $9 + 48 = 57$.
10. **B** (14)
11. **B** (2)

Self-check word **ORDERS**

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · WRITING NUMERICAL EXPRESSIONS

1. a. $7 \times 8 - 10$ b. $50 - 12$ c. $36 \div 9 + 2$ d. $2 \times (15 - 6)$
e. $3 \times 5 + 9$ f. $3 \times (42 \div 7)$
2. $5 \times 6 + 4 \times 3 = 30 + 12 = \42
3. $12 \times 2 + 8 \times 5 - 15 = 24 + 40 - 15 = \49
4. $(8 + 2) \times 5$
5. Sample: 4 friends each buy a \$6 sandwich and a \$3 drink. Total cost $4 \times 9 = \$36$.

SHEET 2 · ORDER OF OPERATIONS WITH POWERS

1. a. 26 b. 50 c. 17 d. 6 e. 20 f. 24
2. a. $(2 + 3) \times 4 = 20$ b. $24 \div (4 + 2) = 4$
3. $3 \times (64 - 4) = 3 \times 60 = \180
4. $5 \times 16 - 30 = 80 - 30 = 50$ beads
5. Lou: add and subtract left to right, $18 - 6 = 12$, $12 + 2 = 14$ (Kim added first).
6. **A** (17)
7. **B** ($2 + 3 \times 4$)